Midterm - ENVS - 193DS

Mark Velthoen

```
library(rstatix)
library(tidyverse)
library(janitor)
library(gt)
#calling in all the necessary packages

tussocks <- read_csv("2016ARFTussockHeight.csv") #creating the tussocks data
frame
temp <- read_csv("temp_2024.csv") #creating the temperature data frame
```

Problem 1. Understanding and critiquing written communication**

a. Null Hypothesis: There is no difference in the mean weight of heated temperature and ambient temperature urchins.

   Alternative Hypothesis: There is a difference in the mean weight of heated temperature and ambient temperature urchins.

b. The researchers ran a student's two-sample t-test. In order to use this test, they must be testing two normally distributed, independent and randomly sampled groups with homogeneous variances and continuous data.

c. t: This is the distribution of the test (Student's t-distribution), which is is a probability distribution used to estimate population parameters when the sample size is small or the population standard deviation is unknown.

   53: This number indicates the degrees of freedom, which relates to the sample size used to estimate variability in the test.

   -13.51: This is the value of the t-statistic, which indicates that there is a significant difference between group means (-13.51 is a large value).

   $p < 0.001$: This is the p-value, which is less than 0.001, signifying that there is a strong difference between the two populations.

   d. One component that could have been included is the significance level. This is an important component to include when addressing the hypothesis because it provides a threshold that allows us to determine whether the difference in mean gonad weight is statistically significant. This is relevant to addressing the hypothesis because it reduces subjectivity and also reduces the risk of a type 1 error.

Another component that could have been included is the sample size for each treatment. Reporting the number of urchins whose gonad weight was measured is important because sample size influences the reliability and precision of the results. A large sample size usually provides a better estimate of mean and reduces the influence of variation.

Problem 2. Interpretation and communication

a. Null Hypothesis: There is no difference in crop value between 2022 and 2023

   Alternative Hypothesis: There is a difference in mean crop value between 2022 and 2023

b. The two pieces of information that are the same are the t-value (-1.5) and the p-value (.14). Two pieces of information that the test outputs includes that the researchers did not include are the degrees of freedom and the confidence interval. The 95% confidence interval provides a range for which if the sampling procedure were repeated numerous times, 95% of the resulting intervals would contain the true population parameter. The paired t-test confidence interval (–51,595,337 to 7,664,551) indicates that the true mean difference in total value between years could range from a decrease of 51.6 million to an increase of 7.7 million, and because the interval includes zero, the change is not statistically significant. The degrees of freedom represent the number of observations in a data set that are free to vary when estimating statistical parameters. The degrees of freedom also indicate how much information is available to calculate variability.

c. Across 28 major crops in Kern County, there was no significant difference in mean crop value between the dry year (2022) and the wet year (2023) following the precipitation whiplash event (Paired t-test: $t(27) = -1.52$, $p= 0.14$).

d. Despite extreme shifts from drought to flooding, Kern County's total crop value remained unchanged ($p = 0.14$). This indicates overall economic resilience. I would advise the blueberry farmer to focus on long term strategies, such as groundwater recharge, as this can help against future volatility.

Problem 3. Reproducing an analysis

a. The response variable is a continuous variable, which is average diameter of a tussock site, in centimeters.

   The predictor variable is categorical, with two groups, un-burned and burned sites.

   The researchers randomly sampled along four 100-m transects, and for each tussock intersecting the transect, the diameter was measured in two directions. (found description of their processes in Methods section, part Micro-topography)

b. Null Hypothesis: There is no difference in mean tussock diameter between the burned and un-burned sites.

   Alternative Hypothesis: There is a significant difference between mean tussock diameter between the burned and un-burned sites.

```
tussocks_clean <- tussocks |> #making a new data framw tussocks_clean
  clean_names() |> #cleaning the variable names
  mutate(site = case_when(site == "Sev" ~ "Burned", #changing the names to
burned and unburned
                          site == "Unb" ~ "Unburned")) |>
  select(site, average_diameter) #selecting only the site and average
diameter variables

slice_sample(tussocks_clean,
             n = 10) #displaying 10 observations from the tussocks_clean data
frame
```

```
# A tibble: 10 × 2
   site     average_diameter
   <chr>               <dbl>
 1 Burned               29.1
 2 Unburned             20.6
 3 Unburned             17.2
 4 Unburned             26.5
 5 Unburned             30.7
 6 Unburned             28.8
 7 Burned               17.8
 8 Unburned             27.0
 9 Unburned             22.7
10 Burned               31.8
```

d.

```
tussocks_summary <- tussocks_clean |> #creating a new data fram
tussocks_summary
  group_by(site) |> #group the data by each site
  summarize(mean = mean(average_diameter), #calculating the mean of average
tussock diameter
            se = sd(average_diameter/sqrt(length(average_diameter)))) |>
#calculating the standard error of average tussock diameter
mutate(mean = round(mean, 1), #rounding the mean and standard error values
       se = round(se,1))
tussocks_summary #displaying the tussocks_summary data table
```

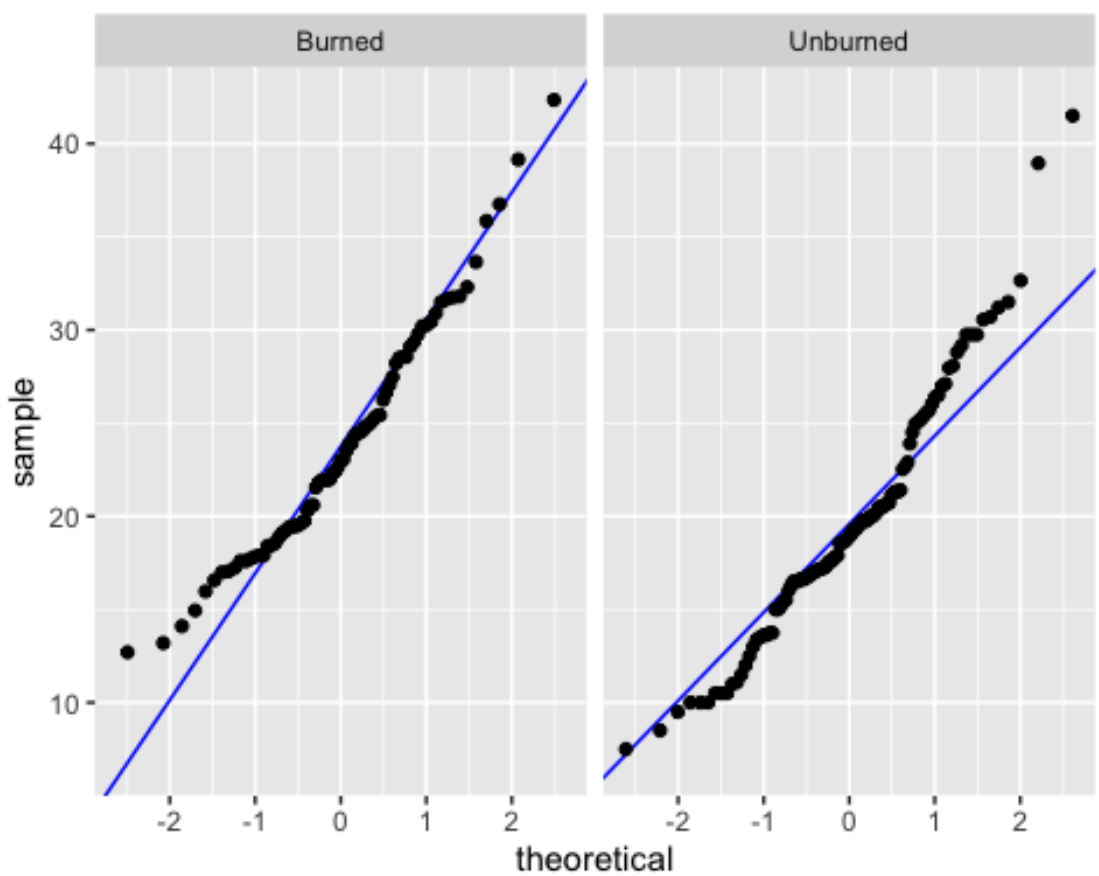
```
# A tibble: 2 × 3
  site      mean    se
  <chr>    <dbl> <dbl>
1 Burned    23.8   0.7
2 Unburned  19.7   0.6
```

```
tussocks_summary |> # using the tussocks_summary data frame for polished
table
select(site, mean, se) |> #selecting the site, mean and standard error
variables
gt() |> # using gt package
cols_label( site = "Site",
mean = "Mean",
se = "Standard error") #labeling the columns correctly
```

Site	Mean	Standard error
Burned	23.8	0.7
Unburned	19.7	0.6

e.

```
ggplot(data = tussocks_clean,  #using tussocks clean dataframe to create a
plot
       aes(sample = average_diameter)) + #y-axis for qqplot is average
diameter
  geom_qq_line(color = "blue") + #showing reference line in blue
  geom_qq() + # adding our observations
  facet_wrap(~site) #show sites seperately
```



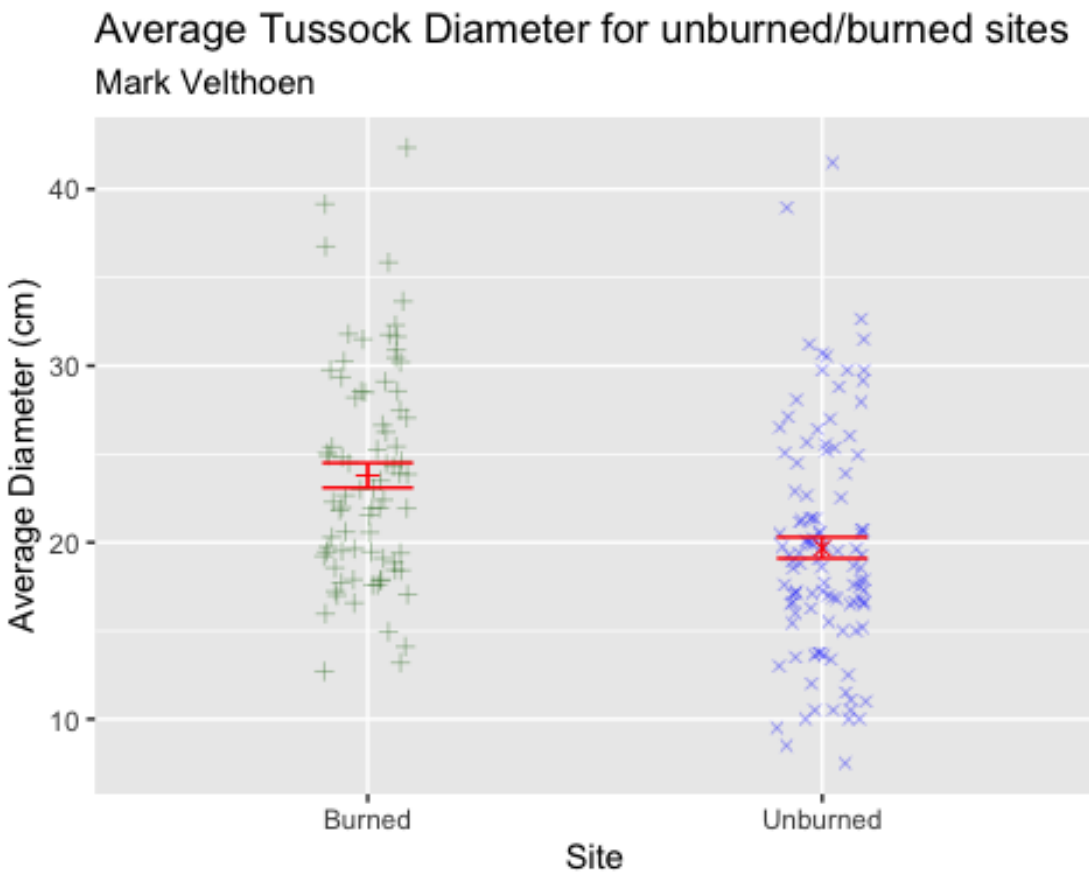
```r
var.test(average_diameter ~ site, #variance test for each site
         data = tussocks_clean) #using tussocks_clean dataframe


    F test to compare two variances

data:  average_diameter by site
F = 0.92558, num df = 78, denom df = 110, p-value = 0.7221
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.6169823 1.4090774
sample estimates:
ratio of variances
         0.9255843
```

After checking my assumptions of normality and homogeneous variances (using a qq-plot and F-test respectively), both assumptions are valid. The qq-plot shows the data (average tussock diameter) for each site plotted against a reference line. Both sites seem to follow the reference line relatively closely, indicating normality. The F-test displays both the p-value (0.72) and the ratio of variances (0.93). Because the p-value is high and the ratio of variances is close to 1, homogeneous variance is verified.

```r
t.test(average_diameter ~ site, #t-test for each site
       var.equal = TRUE, #making it a two sample t-test
         data = tussocks_clean) #using tussocks_clean dataframe


    Two Sample t-test

data:  average_diameter by site
t = 4.4663, df = 188, p-value = 1.371e-05
alternative hypothesis: true difference in means between group Burned and
group Unburned is not equal to 0
95 percent confidence interval:
 2.298075 5.934027
sample estimates:
  mean in group Burned mean in group Unburned
              23.78632               19.67027

tussocks_clean %>% cohens_d(average_diameter ~ site, var.equal = TRUE)
#calculating the effect size using the cohen's d function, with average
diameter as a factor

# A tibble: 1 × 7
  .y.              group1 group2   effsize    n1    n2 magnitude
* <chr>            <chr>  <chr>      <dbl> <int> <int> <ord>
1 average_diameter Burned Unburned   0.657    79   111 moderate
```

g. We measured 79 burned sites and 111 unburned sites and found that burned sites had a larger average tussock mound diameters by about 4 cm (23.79 ± 0.7 cm vs. 19.67 ± 0.6 cm), and this difference was statistically significant (two-sample t-test, $t(188) = 4.47$, $p = 1.37 \times 10^{-5}$), with a 95% confidence interval for the mean difference of 2.30 to 5.93 cm. The effect size was moderate (Cohen's d = 0.66), indicating a meaningful increase in mound diameter at burned sites.

```
ggplot(data = tussocks_clean, #creating a plot using the tussocks_clean data
frame
       aes(x = site, #making the x-axis site type
           y = average_diameter,  #making the y-axis
           color = site,
           shape = site)) + #changing the color and shape for each site
  geom_jitter(height = 0, #making sure to not jitter vertically
              width = 0.1, #jittering the points horizontally
              alpha = 0.3) + #making the points transparent
  geom_errorbar(data = tussocks_summary, #creating an errorbar with
tussocks_summary data frame
                aes(y = NULL, #assign NULL to y
                    ymin = mean - se, #creating the min and max standard
error bars
                    ymax = mean + se),
                width = 0.2, #setting the width
                color = "red") + #coloring the errorbar red
   stat_summary(geom = "point", #using stat summary to create a mean point
               fun = mean,
               color = "red", #coloring point red
               size = 2) + #setting the size
  scale_color_manual(values = c("Burned" = "darkgreen", #manually coloring
the data points for each site
                      "Unburned" = "blue")) +
  scale_shape_manual(values = c("Burned" = 3,
                                "Unburned" = 4)) + #changing the shape for
the points based on site type
  labs(title = "Average Tussock Diameter for unburned/burned sites",
#creating the correct title
       y = "Average Diameter (cm)", #creating the correct x and y axis labels
       x = "Site",
       subtitle = "Mark Velthoen") + #adding my name as the subtitle
 theme(legend.position = "none")
```



i. This figure (figure 3b), titled Average Tussock Diameter for unburned/burned sites, shows a jitter-plot containing points of average tussock diameter (cm) for burned and un-burned sites. The red points indicate the group mean for each site and the error bars represent +/- 1 standard error. Each green + indicates an average tussock diameter observation for burned sites. Each blue x represents an average tussock diameter for unburned sites.

Tussock height and diameter in moist acidic tussock tundra at the site of the 2007 Anaktuvuk River fire scar, and nearby unburned tundra measured in 2016 ver 1. Environmental Data Initiative. https://doi.org/10.6073/pasta/1dccd3fdb3aa693f9c2b69a24f8306ed.

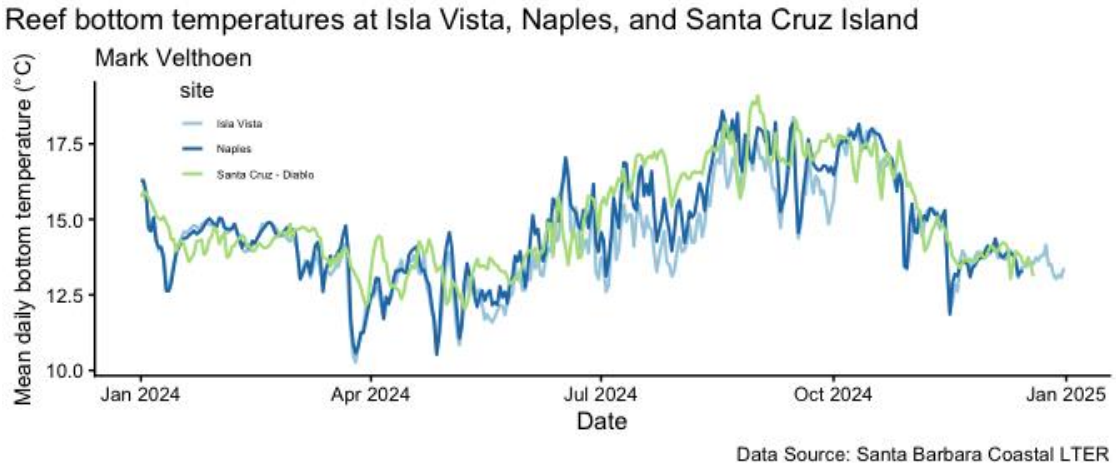
Problem 4. Cleaning, wrangling and visualization

```
temp_clean <- temp |>

  # 1. what package is this function from?
  # This function is from the janitor package
  # 2. What changes after this function and how?
  # After calling this function, all the variables in a data set are
formatted a specific way. All uppercase letters are made lowercase and all
```

```
spaces are replaced with an underscore.
  clean_names() |>

  # 3. what package is year() from?
  # This function is from the lubridate package.
  # 4. what is this combination of mutate() and year() doing?
  # This combination of functions allows us to easily deal with a vast array
of dates. The year function inside of the mutate acts as a way to change the
date_local variable into an integer and basically create a timeline which we
can place our data set onto. The mutate function allows us to add a new
column called year to your dataset that contains only the year extracted from
the date_local dates.
  mutate(year = year(date_local)) |>

  # 5. what package is filter() from?
  # This function is from dplyr.
  # 6. what is being filtered in this functional call?
  # This functional call is filtering to only include the sites of Isla
Vista, Naples, and Santa Cruz Island. The functional call is also filtering
to only include the year of 2024.
  # 7. what sites and year(s) do you expect to be included in the resulting
data frame?
  # I expect that only data from 2024 for the sites of Isla Vista, Naples,
and Santa Cruz Island will be included in the resulting data frame.
  filter(site %in% c("NAPL", "SCDI", "IVEE") &
           year == "2024") |>

  # 8. what column is manipulated?
  # the site column
  # 9. what changes about the values in this column?
  # the site acronyms are changed to full names
  mutate(site = case_when(
    site == "IVEE" ~ "Isla Vista",
    site == "NAPL" ~ "Naples",
    site == "SCDI" ~ "Santa Cruz - Diablo"
  )) |>

  # 10. what is being calculated? and at what scale (e.g. annual, monthly,
daily)?
  # The mean temperature of each site is being calculated for each day.
  # 11. give an example
  # For example, in Isla Vista on January 1st 2024, the mean temperature was
16.3 degrees celcius.
  group_by(site, date_local) |>
  summarize(mean_daily = mean(temp_c)) |>
  ungroup()

slice_sample(temp_clean,
             n = 5)
```

```
# A tibble: 5 × 3
  site                date_local mean_daily
  <chr>               <date>          <dbl>
1 Isla Vista          2024-10-23       16.9
2 Isla Vista          2024-08-23       16.8
3 Santa Cruz - Diablo 2024-05-13       13.7
4 Isla Vista          2024-08-17       16.9
5 Santa Cruz - Diablo 2024-10-21       16.8
```

```
ggplot(data = temp_clean, # creating a plot using temp_clean data frame
  aes(x = date_local, #setting the x-axis to a timeline
      y = mean_daily, #setting the y-axis to mean daily temperature
      color = site)) + #coloring based on site
geom_line(linewidth = 0.7) + #creating the trend line
  labs(x = "Date", #labeling x and y axis
       y = expression("Mean daily bottom temperature ("*degree*"C)"),
       subtitle = "Mark Velthoen",
       title = "Reef bottom temperatures at Isla Vista, Naples, and Santa
Cruz Island", #creating the correct title
       caption = "Data Source: Santa Barbara Coastal LTER") + #adding caption
for data source
  scale_color_brewer(palette = "Paired") + #coloring trendlines using paired
palette
  theme_classic() + #setting the theme
  theme(legend.position = "inside",
        legend.position.inside = c(0.2, 0.8)) + #set the legend position
inside
  theme(plot.title = element_text(hjust = -0.9), # adjusting the location of
the title
        axis.title.y = element_text(size = 10), #changing the size of the y-
axis
        legend.position = c(0.15, 0.83), #positioning the legend
        legend.text = element_text(size = 5), #sizing the legend
    legend.key.width = unit(0.4, "cm"), #sizing the legend key
    legend.key.height = unit(0.4, "cm"),
    legend.title = element_text(size = 10))
```



c. The highest temperatures occurred during August-September 2024. The lowest temperatures occurred during April 2024. The troughs (large dips) on the plotted line indicate the lowest temperatures. The peaks (large spikes) on the plotted line indicate the highest temperatures.